



Benefits of mycorrhizal inoculation to ecological restoration depend on plant functional type, restoration context and time

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ABSTRACT

Mycorrhizal inoculation can enhance outcomes of ecological restoration, but the benefits may be context-dependent. Here, we performed a meta-analysis of field studies to elucidate conditions in which adding mycorrhizal fungi enhances restoration success. We found inoculation increased plant biomass by an average effect size of 1.7 in 70 independent comparisons from 26 field-based studies, with the largest increases to N-fixing woody plants, C4-grasses and plants growing in soils with low plant-available P. Growth responses to inoculation increased with time for the first 3 yr after inoculation, especially for N-fixing woody plants and plants growing in severely altered soils. We found that mycorrhizal inoculation increased species richness of restored plant communities by 30%, promoted establishment of target species, and enhanced similarity of restored to reference communities. We conclude that the addition of mycorrhizal fungi to restoration sites can facilitate rapid establishment of vegetation cover, and restoration of diverse plant communities more akin to reference sites.

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1. Introduction

Human activities have degraded nearly two-thirds of all ecosystems (Millennium Ecosystem Assessment, 2005) with significant consequences for biological diversity, ecosystem functioning and provisioning of ecosystem services (TEEB, 2010). In many cases, degraded ecosystems need ecological restoration (SER, 2004), but achieving recovery of ecosystem function and diversity can take decades or may ultimately fail due to abiotic (e.g. eutrophication, acidification) and biotic constraints (e.g. competition, unfavourable soil community composition) (Suding et al., 2004; Grman et al., 2015). Identifying factors that limit effective ecological restoration, and management approaches to overcoming these factors, are

key steps towards improving restoration outcomes (Hobbs and Harris, 2001; Harris, 2009; Perring et al., 2015). Experiments have demonstrated the central role of soil biota and soil processes in plant community development and ecosystem functioning (Kardol et al., 2006; Carbajo et al., 2011; Van der Putten et al., 2013), implying that manipulation of the soil community could be an important driver of restoration success in terrestrial ecosystems (Wubs et al., 2016).

Mycorrhizal fungi in particular play a critical role in recovery of degraded terrestrial ecosystems because plants benefit from hosting these obligate symbionts (Kardol and Wardle, 2010). Indeed, mycorrhizal fungi are ubiquitous in ecosystems, occurring in roots of 80–90% of all vascular plant species (Brundrett, 2009). In exchange for plant-assimilated carbon, mycorrhizal fungi can enhance plant nutrient supply (mainly N and P) (Smith and Read, 2008), alleviate plant abiotic stress (e.g. drought stress; Augé et al., 2015) and increase plant-resistance to pathogens (Jung

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et al., 2012). Through these mechanisms, they can directly benefit plant performance, and indirectly affect plant community structure by mediating biotic interactions (Moora and Zobel, 2010; Klironomos et al., 2011). Further, extra-radical hyphae and glomalin production by mycorrhizal fungi can improve soil physical characteristics, particularly soil aggregate stability and resistance to soil erosion (Rillig et al., 2002).

Notwithstanding these potential benefits, recent meta-analyses have revealed that effects of mycorrhizal inoculation on plant growth are context dependent; suggesting they vary with host plant characteristics, soil abiotic and biotic characteristics, sources of fungal inocula and experimental procedures. For example, effects of mycorrhizal inoculation on plant growth (Hoeksema et al., 2010; Maltz and Treseder, 2015), plant competitive ability (Lin et al., 2015) and plant invasion success (Bunn et al., 2015) depend on plant functional group. While these findings suggest context dependency of mycorrhizal inoculation effects, their scope of inference for ecological restoration of plant communities in field conditions is limited because the data used in these meta-analyses originated from inoculation experiments under controlled conditions (i.e. predominantly pot experiments; Hoeksema et al., 2010; Bunn et al., 2015; Lin et al., 2015; but see Maltz and Treseder, 2015). Field experiments are crucial to predicting the effectiveness of mycorrhizal inoculation in ecological restoration because outcomes depend on soil legacies associated with land use history (e.g. Standish et al., 2009). In addition, pot experiments may show that mycorrhizal fungi are potentially beneficial for plants under field conditions, but do not indicate whether inoculation is required in the field. Mycorrhizal fungi may be present at a restoration site, or able to disperse to it, and colonise plant roots as they establish (Kulmatiski and Beard, 2011; Garcia de Leon et al., 2016). Consequently, understanding the context dependence of the effectiveness of mycorrhizal fungi in ecological restoration requires greater emphasis on field conditions.

Plant community composition and diversity are often measures of restoration success, yet the primary response variable in most studies has been plant growth (Harris, 2009; Kardol and Wardle, 2010). Plant composition commonly differs markedly between reference and restored grassland communities, typically with lower plant diversity and lower proportions of some plant functional groups in restored sites (Kindscher and Tieszen, 1998; Piper et al., 2007; Kiehl et al., 2010). Reference sites usually refer to the historic, pre-disturbance state of the ecosystem that serve as a model for setting and achieving restoration goals (SER, 2004). Some authors have hypothesised that absence of mycorrhizal fungi, and the subsequent competitive disadvantage of plant species or functional groups that strongly rely on mycorrhizal symbiosis (e.g. legumes; Wilson and Hartnett, 1998; Fitter, 2005; Koziol and Bever, 2015; Koziol and Bever, 2016), might explain the differences in composition of reference and restored plant communities. This hypothesis, however, remains poorly tested (but see Moeslund et al., 2017). Since limited establishment of sown target species still often challenges restoration practitioners and restricts recovery of whole plant communities (e.g. Grman et al., 2015), more research is needed to understand the potential for mycorrhizal fungi to promote plant diversity and composition more akin to reference sites (cf. Barea et al., 2011).

We conducted a meta-analysis of the context dependency of plant responses to mycorrhizal inoculation in field-based restoration projects, focusing on plant growth (biomass and height) and plant species richness as the response variables. Using 34 experimental studies of woody and grassy ecosystems (26 for plant biomass; 11 for plant height; five for plant richness), from five continents reporting one or more of these response variables, we assessed the effects of six explanatory factors related to: plant

identity (plant functional group), soil conditions (soil pH, concentrations of total N and plant-available P), disturbance history (severity of soil disturbance), fungal identity (inoculum complexity), and experimental conditions (control contrast). Further, given the importance of time as a factor in success of restoration projects (Harris, 2009) we assessed the development of inoculation effects over time and the context dependency of these time effects.

2. Materials and methods

2.1. Literature search and dataset construction

To select relevant studies for our meta-analysis, we searched the ISI Web of Science database (1900–2016) using the keyword string *mycorrhiza** AND (restoration OR reclamation OR rehabilitation). This search identified 758 studies.

We reviewed the abstracts of the 758 studies and selected the subset of studies that tested the effect of mycorrhizal inoculation on plant performance in the context of ecological restoration, which reduced the number of studies to 380. Of these 380 studies, we included in the meta-analysis those studies, which experimentally added a mycorrhizal inoculum (arbuscular mycorrhizal fungi and/or ectomycorrhizal fungi) and compared inoculation effects with a non-inoculated control contrast under field conditions. We included only those studies in our meta-analysis that used gamma irradiation or autoclaving for soil sterilization, and excluded those that applied fungicides, since the latter does not eliminate all fungi, but rather shifts fungal community composition, and thus biases results in control plots. Further, studies were included in the meta-analysis, if they provided treatment means, a measure of sampling variance (standard deviation or standard error) and sample sizes for measures of plant biomass (whole plant biomass or alternatively shoot biomass), plant height or plant species richness. These restrictions reduced the number of studies to 34, including 26 for the plant biomass response variable, 11 for the plant height response variable and five for the plant species richness response variable (Fig. 1). Other measures of plant species diversity, and cover ratios of different plant groups were also measured, but too infrequently to be included in separate analyses ($n = 1$ study of each measure respectively). Standard deviations were back calculated from standard errors as necessary ($SD = SE \times \sqrt{n}$).

Data were extracted from the text, tables and figures in each publication. From each study, we extracted data on time since restoration, plant functional group, soil conditions (soil pH, concentrations of total N and plant-available P), disturbance history, inoculum complexity and experimental conditions (control contrasts) (Table 1) as potential factors influencing mycorrhizal effects on plant performance (Hoeksema et al., 2010; Bunn et al., 2015; Lin et al., 2015; Maltz and Treseder, 2015). We assigned levels to categorical factors as follows (see Table 1 for details): The factor 'plant functional group' separated studies according to plant growth forms (woody species, grasses, forbs) and nutrient uptake strategies (N-fixation, root architecture), resulting in six levels. The factor 'disturbance history' had two levels according to the degree of soil disturbance at the restoration site (soil severely altered, soil not altered). The factor 'inoculum complexity' had four levels, according to the number of fungal species in the inoculum (single to whole soil inoculum). Most of the studies (about 80%) used local inocula, the rest applied commercial inoculum. 'Control contrast' was a three-level factor characterising whether inoculation effects had been compared to control soils that had been sterilized and a microbial washing had been applied in the beginning of the experiment. In a restoration context, where the control is no inoculum (biomass: $n = 28$; height: $n = 5$), addition of inoculum

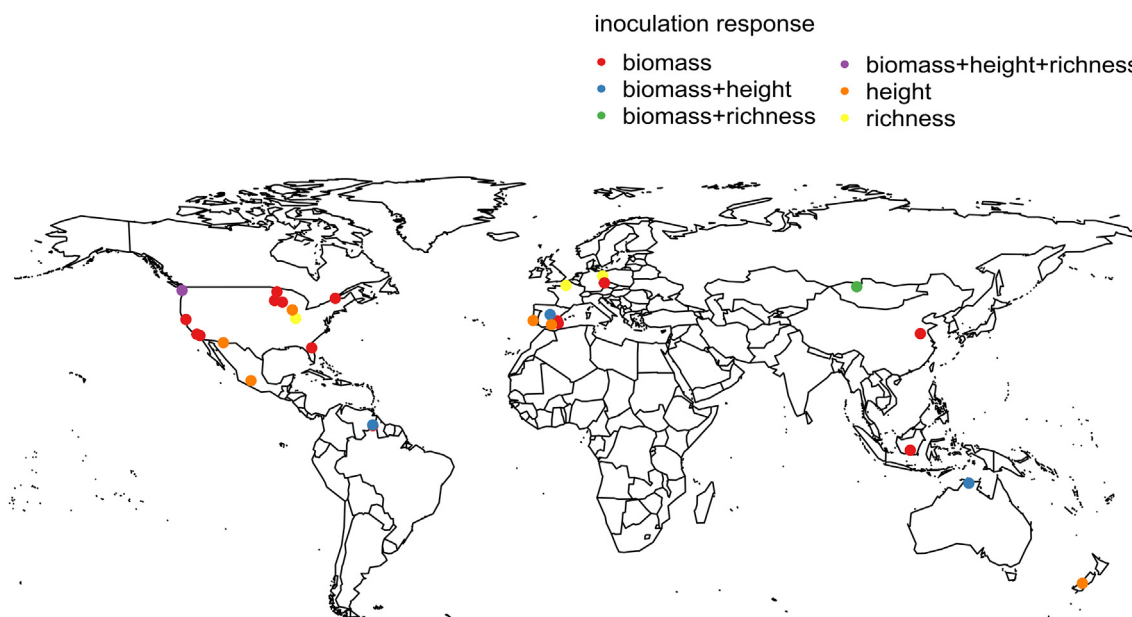


Fig. 1. Map showing the locations of 34 restoration experiments and coloured by plant responses to mycorrhizal inoculation included in the meta-analysis. Number of locations ($n = 29$) is smaller than the number of studies as some studies were located at the same place. Between 13 and 163 comparisons of plant biomass, plant height and plant species richness were available using these restoration studies.

Table 1

Explanatory factors used to assess the context dependency of inoculation effects on plant biomass, height and richness.

Explanatory factors	Categories/units	Proportion of comparisons where data was available (%) for plant biomass (global dataset, time dataset)	Proportion of comparisons where data was available (%) for plant height (global dataset, time dataset)	Proportion of comparisons where data was available (%) for plant richness
Time (since inoculation)	number of months	100, 100	100, 100	100
Plant functional group	C3-grass, C4-grass, non-N-fixing forb, forb-grass mix, non-N-fixing woody, N-fixing woody	100, 100	100, 100	100
Habitat conditions				
pH	1–14 (min = 4.4; max = 10.2)	67, 100	87, 100	44
soil total N	g kg^{-1} (min = 0; max = 12)	60, 100	67, 100	11
plant-available P	mg kg^{-1} (min = 0; max = 61)	63, 91	87, 91	44
Disturbance history ^a	soil severely altered, soil not altered	100, 100	100, 100	100
Inoculum complexity ^b	single, mix, mix-whole soil, whole soil inoculum	100, 100	100, 100	100
Control contrast	control = no inoculum ^c , control = sterile soil, control = sterile soil & microbial wash	100, 100	100, 100	100

^a Categories of disturbance history reflect the degree of soil disturbance. “Soil severely altered” included studies in which disturbance activities removed or severely altered the topsoil (e.g. mining, road construction, tree logging with bulldozers, agriculture), assuming that mycorrhizal fungal communities had been severely altered by these activities. Otherwise, studies were categorized as “soil not altered”.

^b Categories of inoculum complexity were chosen as follows: Single inoculum is an inoculum containing a single fungal taxon originating from commercial inoculum or natural soil applied as a tablet, water-mixture or solution of water and nurse plant roots. A mix inoculum is a mixture of 2–10 fungal taxa originating from commercial inoculum or natural soil applied as a tablet, water-mixture or solution of water and nurse plant roots. Mix-whole soil inoculum refers to studies comparing effects of inoculation with a fungal mix to whole soil inocula using the same control for both treatments. Thus, comparisons are not statistically independent and effect sizes were calculated as the average effect of both types of inocula. Whole soil inoculum consists of soil collected from reference sites, expected to contain the complete biotic soil community i.e. organisms including but not exclusive to mycorrhizal fungi. All studies comparing fungal mixtures to whole soil inocula used AM fungi. For AM fungi, inoculation effects stemming from whole soil inocula can be related to mycorrhizal fungal effects, if they are compared to a control, which has received a microbial filtrate (wash) containing the majority of soil organisms of the experimental soil, except AM fungi (pore size 50 μm ; Koide and Li, 1989). This method works effectively for AM fungi due to their relatively large spores compared to the majority of other soil microbial communities (e.g. bacteria and non-AM fungi) (Uibopuu et al., 2012). For plant biomass 100% of all comparisons using whole soil inoculum and 50% of all comparisons using the “mix-whole soil” type of inoculum soil applied a microbial wash to control treatments, 50% applied no specific control treatment (no inoculum). For plant height, 100% of the comparisons using whole soil inoculum or the “mix-whole soil” type inoculum applied microbial wash to control treatments.

^c No inoculum refers to the type of control treatment where no specific mycorrhizal control treatment was applied, but plants were grown in untreated soil at the restoration site. Sterile soil refers to the type of control treatment where control plants were grown in field soil that had been sterilized by radiation or autoclaving. Sterile soil plus microbial wash refers to the type of control treatment where a microbial filtrate (wash) containing the majority of soil organisms of the experimental soil, except mycorrhizal fungi, was added to the sterilized control soil.

reflects expected benefits for ecological restoration, assuming that a proportion of native fungi and bacteria are already present in the soil. In contrast, where the control is sterilized soil, to exclude effects of the local soil biota, with or without microbial wash added (biomass: $n = 42$; height: $n = 24$), addition of inoculum does not reflect expected benefits for ecological restoration, because the role of mycorrhiza that may have occurred at the restoration site is not accounted for in the design.

We included multiple individual comparisons from experimental studies e.g. from different study sites, host plants, inoculation treatments and multiple time points. This resulted in 163 comparisons for plant biomass, 97 comparisons for plant height and 13 comparisons for plant species richness, covering 14 countries and five continents (Fig. 1). We created five datasets for analysis from these data: two for plant biomass, two for plant height and one for plant richness (number of plant species per study plot). The two different plant biomass and plant height datasets were designed to assess (1) the general effect of inoculation on plant biomass including the effect of different explanatory factors (hereafter referred to as 'global dataset') and (2) changes in inoculation effects over time up to 36 months after inoculation (hereafter referred to as 'time dataset'). The two global datasets comprised all comparisons, but included only data from the last harvest for studies that reported plant responses to inoculation over multiple points in time (Table 2). The two time datasets comprised all studies of the global dataset that reported plant response to inoculation over multiple points in time including comparisons of all points in time (Table 2). There were not enough studies to assess the effect of time on response of plant richness to inoculation ($n = 1$). Further details for each dataset (number of studies, comparisons, continents and countries, and vegetation types) are shown in Table 2 and Supplementary Material S1.

2.2. Data analysis

Effect sizes of plant response to mycorrhizal inoculation were calculated as the natural logarithm of the response ratio ($\ln(R) = \ln(\text{inoculated/non-inoculated})$) after Borenstein (2009) and similar to Hoeksema et al. (2010) with which we compared our results. Moreover, this metric is symmetric around zero, and it is positive when there is a beneficial effect of inoculation, and negative for a detrimental effect, which aids interpretation of results. To guarantee the independence of experimental comparisons, we calculated the combined effect size and sample variance for comparisons from the same study that shared one control (Borenstein, 2009). One comparison had an unusually high effect size and variance, and thus was omitted from the analyses (Table S2, comparison marked with*). See Table 2 for details on the datasets (number of comparisons, studies) and Table S2–S6 for the complete datasets.

Random mixed-effects models were used to estimate the size and significance of mycorrhizal inoculation on plant biomass, height and richness, as well as the effects of explanatory factors (as listed in Table 1) on these plant responses to mycorrhizal

inoculation. Inoculation effects were calculated as mean effect size weighted by sample variance (Viechtenbauer, 2010), and were interpreted as significant when the weighted mean and its 95% confidence intervals (CI) differed from zero (Borenstein, 2009). The effect of time on plant response to inoculation (within studies) was modelled including time after inoculation (in months) as a fixed effect. The interactive effects of explanatory factors and time were also modelled. In all models, study was included as a random factor to account for methodological differences among studies. For models of the time datasets, the random factor treatment-nested-within-study was included where there were multiple comparisons of independent experimental treatments (e.g. control contrasts, host plants) from the same study site. All models were fitted using restricted maximum-likelihood estimation (REML) in the *metafor* package (Viechtenbauer, 2010) in R (3.1.2. R Core Team, 2014).

To test for potential publication bias we assessed the results of each dataset visually via funnel plots (Borenstein, 2009) and tested for asymmetry using Egger's regression test (function *regtest* in the *metafor* package), but did not find any indication of publication bias, i.e. an overrepresentation of studies reporting positive inoculation effects. However, it is possible that some negative or insignificant results of restoration experiments have not been published.

3. Results

3.1. Plant response to mycorrhizal inoculation and its context dependency

Mycorrhizal inoculation increased plant biomass by an average of 1.7, and plant height by an average of 1.3 compared with controls (Fig. 2 and Fig. S8). Factors significantly influencing inoculation effects were similar for plant biomass and plant height, so we focus here on biomass results (Table S7 and Fig. S8 for plant height results).

Plant functional group significantly influenced the magnitude of inoculation effects (Fig. 2). Inoculation significantly increased biomass of C3- and C4-grasses, N-fixing and non-N-fixing woody species and mixtures of forbs and grasses, but had no effect on non N-fixing forbs (Fig. 2). The response of plant biomass to inoculation was about two fold larger for C4- compared with C3-grasses, and for N-fixing compared with non-N-fixing woody species (Fig. 2). For soil conditions, higher concentrations of total N and plant-available P significantly, yet weakly, reduced benefits of inoculation on average by 0.03 per g kg⁻¹ total N and -0.01 per mg kg⁻¹ plant-available P (Fig. 3). Soil pH did not affect plant biomass response to inoculation ($y = 0.27 + 0.04x$; $p = 0.313$). Disturbance history had no significant effect on inoculation effects, when considering the last harvest of each experiment (global dataset, Fig. 2). However, the time dataset demonstrated significantly larger inoculation effects on plant biomass in severely altered soils compared with soils that were not altered (Fig. 4C).

There were no significant differences in the strength of

Table 2
Attributes of the five datasets considered in the meta-analyses of the 34 restoration experiments mapped on Fig. 1.

Response variable	Dataset	Number of					Vegetation
		studies	comparisons (total)	independent comparisons	continents	countries	
Plant biomass	Global dataset	26	125	70	5	9	woody, grassland
Plant biomass	Time dataset	8	60	35	3	4	woody, grassland
Plant height	Global dataset	11	64	29	5	7	woody, grassland
Plant height	Time dataset	6	54	32	3	4	woody, grassland
Plant richness	Global dataset	5	13	9	3	4	grassland

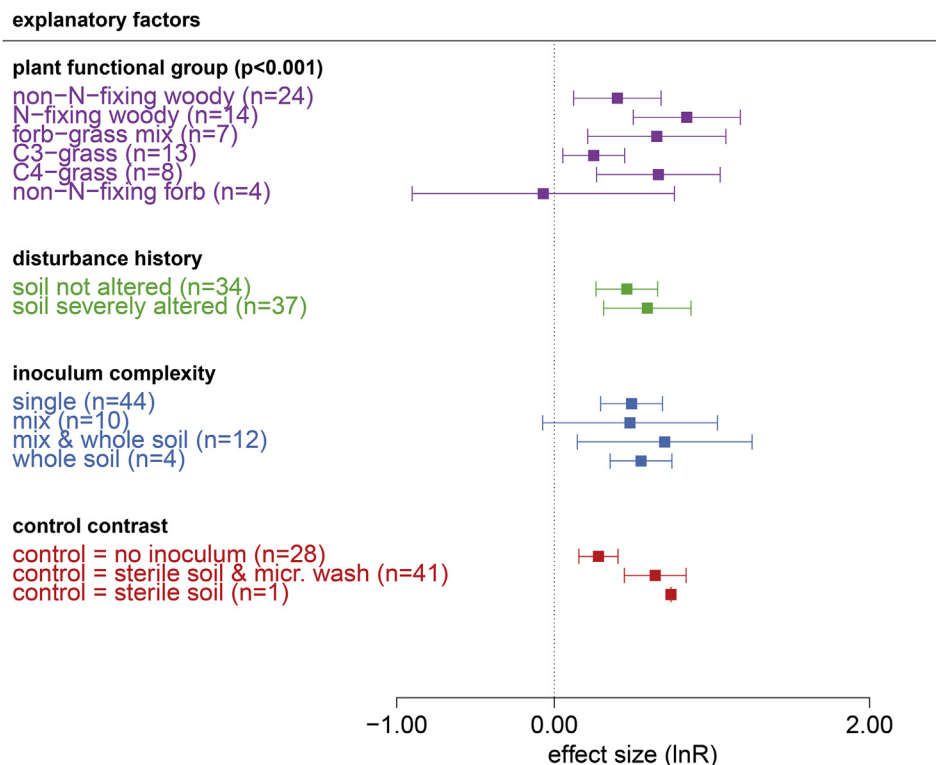


Fig. 2. Plant biomass response to inoculation for the global dataset ($n = 70$), and levels of categorical explanatory factors 'plant functional group', 'disturbance history', 'inoculum complexity' and 'control contrast'. Symbols are means (closed squares, centre of the diamond) $\pm$ 95% confidence intervals (error bars, right/left tip of the diamond). Differences among factor levels were estimated according to random-effects models. Effect size is the natural logarithm of the response ratio where a positive effect size equates to a benefit of inoculation on plant biomass.

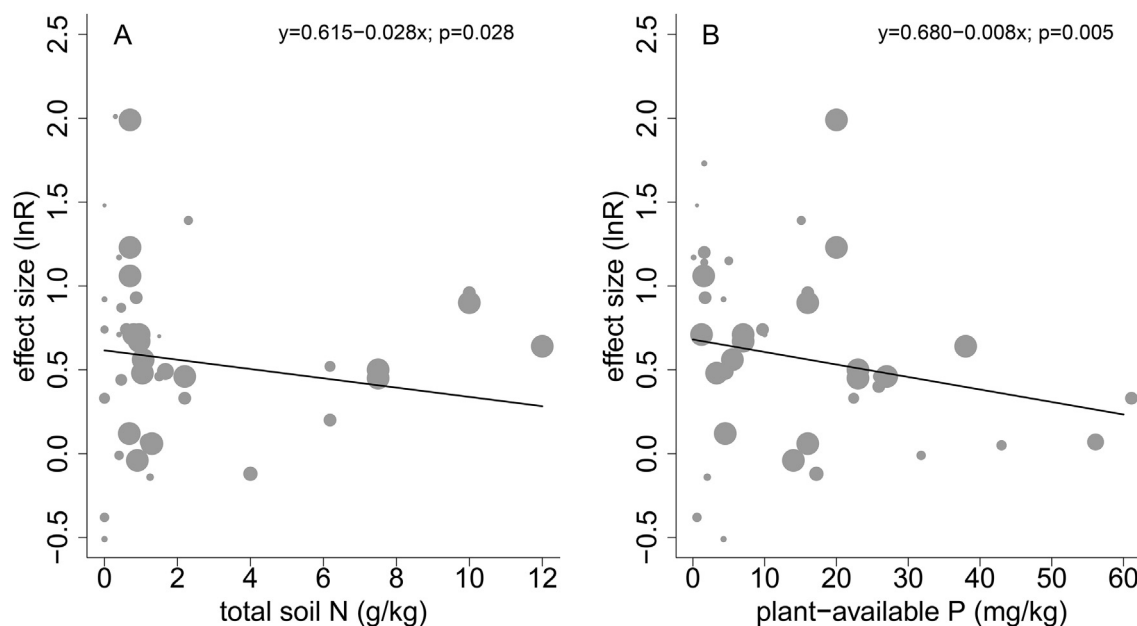


Fig. 3. Relationships of plant biomass response to inoculation and increasing concentration of (A) total N and (B) plant-available P in the soil ($N: n = 42$; $P: n = 44$). Effect size is the natural logarithm of the response ratio where a positive effect size equates to benefit of inoculation on plant biomass. Solid lines indicate significant relationships between explanatory factors and plant biomass response. Symbol sizes are proportional to the inverse of the standard errors (i.e. larger/more precise studies are represented by larger dots).

inoculation effects in relation to inoculum complexity and type of control contrast (Fig. 2). Yet, inoculation effects on plant biomass were larger in studies comparing whole soil inocula and mixes (mix and whole soil) to other inoculation types (Figs. 2 and 4D). Whole

soil inoculum consists of microorganisms other than mycorrhizal fungi, but this was controlled for in the majority of cases through use of microbial wash (with mycorrhizal fungi removed) as control contrast (Table 1^{b,c}; Uibopuu et al., 2012). Inoculation effects tended

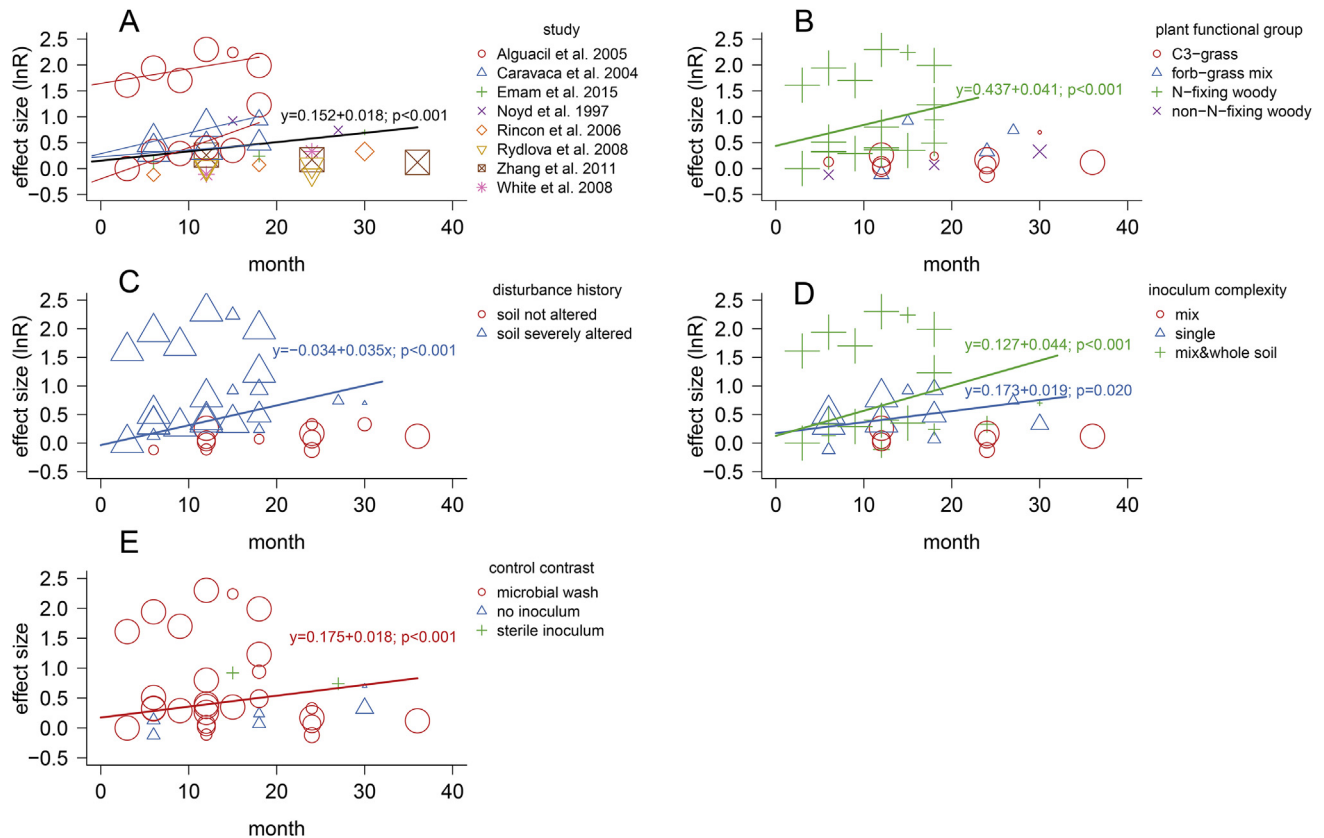


Fig. 4. Relationship between time (i.e. months after inoculation) and plant biomass response to inoculation (i.e., effect sizes as natural log of response ratio where positive values equate to benefit of inoculation on plant biomass) for different levels of categorical factors: (A) independent time series (within studies) and overall effect, (B) plant functional group, (C) disturbance history, (D) inoculum complexity and (E) control contrast. Only explanatory factors with significant variation in time-effects among factor levels are shown. Solid lines indicate significant relationships between time and plant biomass response. Symbol sizes are proportional to the inverse of the standard errors (i.e. larger/more precise studies are shown as larger symbols). $n = 35$.

to be larger in experiments using sterile soil and microbial wash as the control contrast compared with experiments using non-inoculated reference soil (Fig. 2).

3.2. Effect of time on plant response to mycorrhizal inoculation and its context dependence

The random-effects models indicated that inoculation effects on plant biomass increased weakly, but significantly over time. With each month of experimental duration, plant biomass response to mycorrhizal inoculation increased on average by 2% (Fig. 4A). A range of explanatory factors influenced plant biomass response over time affecting the strength of time effects (Fig. 4), but time had a positive effect for most of the experiments. The response of plant height to inoculation over time revealed similar but weaker patterns (see Table S9 for plant height results).

Inoculation effects most strongly increased over time in experiments with N-fixing woody species as target species (Fig. 4B), and in experiments where soil had been severely altered prior to restoration (Fig. 4C). Inoculation effects increased significantly, but moderately, over time in experiments using either single species inoculum or comparing a fungal mixture to whole soil inoculum (Fig. 4D), and in experiments applying sterile soil and microbial wash as control contrast (Fig. 4E). Inoculation effects became significantly weaker over time for soil pH, and increasing soil pH reduced the positive effects of time on plant response to mycorrhizal inoculation (Table 3). Inoculation effects were stable over time for experiments using C3-grasses as host plants (Fig. 4B),

Table 3

Relationship between time (i.e. months after inoculation) and plant biomass response to inoculation (i.e., effect sizes as natural log of response ratio where positive values equate to benefit of inoculation on plant biomass) for different soil nutrient levels: total N (g kg^{-1}), plant-available P (mg kg^{-1}), and pH. Significant effects according to random-effects models are highlighted in bold. $n = 35$ (N, pH), $n = 32$ (P).

explanatory factors	estimate	SE	p-value
time* P	0.0004	0.0002	0.057
time* N	−0.008	0.005	0.106
time*pH	−0.015	0.003	<0.001

revegetating soils not severely altered prior to restoration (Fig. 4C), and using a fungal mixture as inoculum (Fig. 4D). However, explanatory factors were sometimes linked; for example, most of the experiments using N-fixing species, also used a single species inoculum or compared a fungal mixture to a whole soil inoculum.

3.3. Effect of mycorrhizal inoculation on plant species richness and its context dependency

Mycorrhizal inoculation led to an average 1.3-fold increase in plant species richness. We detected no significant effects of explanatory factors on the magnitude or direction of the plant richness response (Fig. 5). Inoculation tended to increase the number of native and specialist species, thereby promoting similarity of the restored to the reference ecosystem (Table 4).

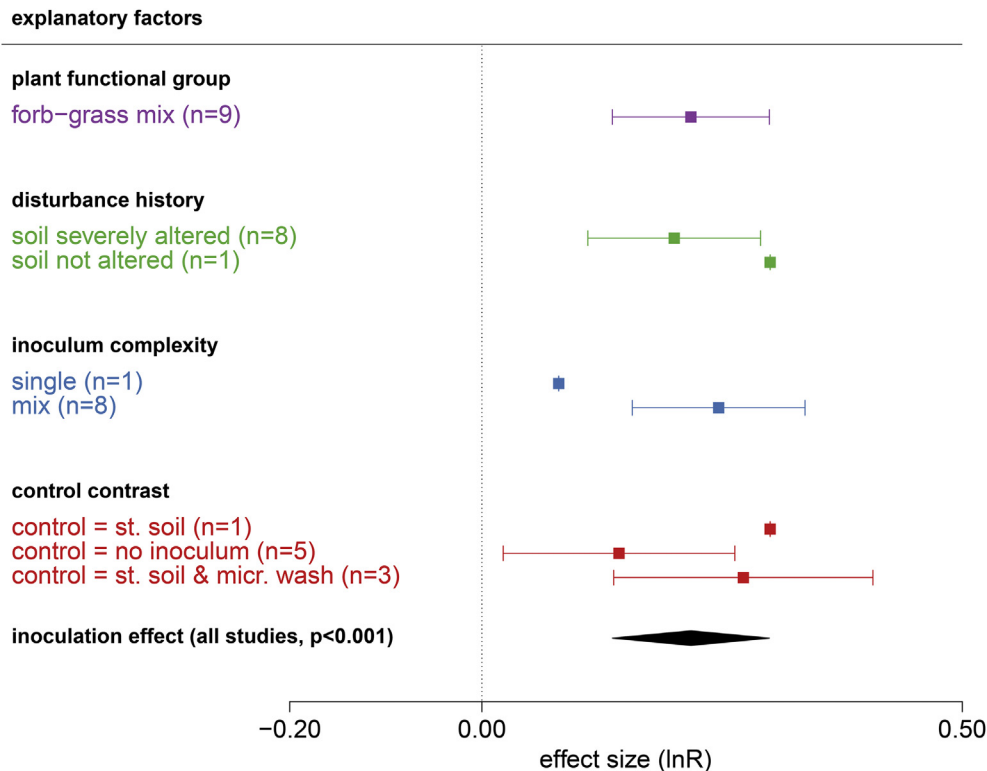


Fig. 5. Plant richness response to inoculation for the whole dataset ($n = 70$), and levels of categorical explanatory factors 'plant functional group', 'disturbance history', 'inoculum complexity' and 'control contrast'. Symbols are means (closed squares, centre of the diamond) $\pm$ 95% confidence intervals (error bars, right/left tip of the diamond). Differences among factor levels were estimated according to random-effects models. Effect size is the natural logarithm of the response ratio where a positive effect size equates to a benefit of inoculation on plant richness.

Table 4

Summary of inoculation effects on plant community composition in restored grasslands. Coding indicates direction and significance of inoculation effects: no effect (0), non-significant positive trend (+), significantly positive (++), no data available (NA). $n = 9$.

Response variable	Cook et al., 2011 (n = 1)	Fischer et al., 2013 (n = 1)	Kozioł and Bever, 2016 (n = 1)	Torrez et al., 2016 (n = 3)	Zhang et al., 2012 (n = 3)
Number of target species	0	+	++	++	++
Number of native species	0	+	++	++	++
Similarity to reference	NA	NA	++	++	NA

4. Discussion

Our meta-analysis of field restoration experiments in grasslands, shrublands and woodlands spanning semi-arid to mesic sites around the world revealed that inoculation with mycorrhizal fungi enhances plant growth (biomass and height). Additionally, for a subset of these studies, plant species richness and similarity of restored species composition to the reference communities were also promoted. The strength of effects of mycorrhizal inoculation on plant growth were strongly context-dependent. In particular, inoculation was most beneficial to plant functional groups with greater nutrient demand (woody N-fixers), plants with inefficient nutrient-uptake (C4-grasses) and plants growing on soils low in total N and plant-available P. The positive effect of inoculation on plant growth increased over timeframes of at least 3 yr, especially for N-fixing woody species. Time series analyses suggested that inoculation effects on plant growth tended to be larger in experiments where soil was severely disturbed prior to restoration, indicating that disturbance history can affect restoration efforts through alteration of the soil mycorrhizal fungal community. Inoculation effects were not significantly influenced by inoculum complexity and type of control contrast.

4.1. Context-dependency of inoculation effects on plant growth

Our results showed that nutrient demand, nutrient use-efficiency and soil P concentration were the top three predictors of plant growth response to mycorrhizal inoculation, as demonstrated by strong growth responses of N-fixing plants and C3 grasses, and in sites with low soil P concentration, respectively. These results match the well-documented role of mycorrhizas in facilitating nutrient-uptake (Koide, 1991; Schwartz and Hoeksema, 1998; Smith and Read, 2008). Greatest increase in biomass of woody N-fixers in response to mycorrhizal inoculation is consistent with the high P-costs for N-fixation fostering plant reliance on mycorrhizal symbiosis for P-uptake (Mortimer et al., 2008; Wagg et al., 2011). The observation of greater benefits of inoculation for C4-, compared with C3- grasses (see also Hoeksema et al., 2010), likely reflects the higher dependency of C4-grasses on mycorrhizal symbiosis for nutrient-uptake due to the inefficient nutrient uptake of their coarse root system (Hetrick et al., 1990; Wilson and Hartnett, 1998). Our finding of a negative effect of high total N and high plant-available P on plant growth response to inoculation is consistent with earlier findings that plants gain more from trading C for fungal-derived nutrients under low rather than high

nutrient availability (Koide, 1991; Schwartz and Hoeksema, 1998; Jones and Smith, 2004). However, the effect of total N and plant-available P in soils on plant response to inoculation was weak, perhaps due in part to variability in the soil nutrient data because of differences in methodology and timing of the measurements of total N and plant-available P (e.g. Mallarino, 1995; Bremer et al., 1996).

The lack of a response of plant growth to the number of fungal taxa in the inoculum (i.e. inoculum complexity) (see also Hoeksema et al., 2010; Maltz and Treseder, 2015) may reflect the low host specificity of AM fungi (Smith and Read, 2008). However, plant growth responses tended to be greater for whole soil inocula, which might be due to their diverse fungal community, providing a large set of complementary functions (Van der Heijden et al., 1998; Hart and Klironomos, 2004), and/or due to better adaptation of native fungal communities to the environmental conditions at the restoration site. Our findings are consistent with earlier studies (Barea et al., 2011; Maltz and Treseder, 2015), highlighting the high effectiveness of whole soil inocula for ecological restoration. Additionally, plant response was greatest when compared with sterile control soils, similar to findings under controlled conditions (Lin et al., 2015), suggesting that higher activity of native mycorrhizal fungal communities in unsterilized control soils reduced the difference between inoculated versus control treatments (e.g. Cook et al., 2011). The unsterilized control soils provide a more accurate estimate of the potential of mycorrhizal fungi to increase restoration success, i.e. benefit of applying a mycorrhizal inoculum over not applying one (e.g. Torrez et al., 2016).

4.2. Effect of time on plant response to mycorrhizal inoculation and its context-dependency

Inoculation effects increased with duration of experiments in the short-term (i.e. up to 3 yr, which is expected given that mycorrhizal fungi need time to populate a site and to establish common mycelial networks (Van der Heijden and Horton, 2009; Van der Putten et al., 2013; Ke et al., 2015). Increase in inoculation effects with time was particularly prominent for studies using woody N-fixers as host plants for initiating revegetation of severely altered soils. Previous studies emphasise the importance of soil mutualists such as mycorrhizal fungi and N-fixing bacteria during early stages of vegetation re-establishment (Reynolds et al., 2003; Kardol et al., 2013; Dudenhöffer et al., 2018). The majority of studies using woody N-fixers for initiating revegetation were conducted in semi-arid regions with low vegetation cover and microbial activity at the restoration sites (Caravaca et al., 2004; Alguacil et al., 2005). Thus, increasing inoculation effects over time may reflect not only increasing mycorrhizal facilitation as mycorrhizas establish, but also the self-promoting changes in microclimate, soil fertility and microbial activity due to the successful establishment of woody N-fixers as nurse species (Barea et al., 2011).

Stronger benefits from inoculation under severe soil disturbance compared with non-altered soil highlights the importance of soil legacies influencing restoration outcomes (Prach and Hobbs, 2008; Standish et al., 2009). For example, reduced fungal abundance and altered fungal composition after severe soil disturbance can hinder natural recovery of the mycorrhizal fungal community (Lekberg and Koide, 2005; Moora et al., 2014). Our findings show that in these cases inoculation with mycorrhizal fungi can assist the recovery of the mycorrhizal fungal community. The lack of an effect of disturbance history in the full dataset may be due to the wide range of disturbance histories and restoration activities included in the category 'severely altered soils'. Further research is needed to examine the longer-term benefits of mycorrhizal inoculation for a diversity of plant functional groups, disturbance histories and

management interventions. In particular, the effects of inoculation on successional trajectories of plant communities is an avenue worthy of exploration (Teste, 2016).

4.3. Inoculation effects on plant richness

Mycorrhizal inoculation promoted plant species richness, and greater similarity in plant species composition to reference communities (Fischer et al., 2013; Koziol and Bever, 2016; Torrez et al., 2016), demonstrating that mycorrhizal inoculation can facilitate restoration of diverse communities more akin to reference sites. Our results match findings from microcosm experiments (Van der Heijden et al., 1998), but to our knowledge, our study is the first to synthesise the emerging body of research examining mycorrhizal benefits on species richness under field conditions (i.e., dating from 2011; Table S5). Positive inoculation effects on target community composition may reflect the balancing effect of mycorrhizal fungi on plant inter-specific competition (Moora and Zobel, 1995), i.e. increasing the biomass of competitively subordinate species compared to that of dominants (Grime et al., 1987). For example, addition of mycorrhizal fungi can increase the competitive ability of seedlings of sown mycorrhizal-fungal-dependent target plants, which might be inferior competitors in a situation when mycorrhizal fungal inoculum is lacking or low in abundance (Torrez et al., 2016). Thus, mycorrhizal inoculation may facilitate restoration of diverse communities by promoting seedling establishment of sown target species in restoration sites—a challenge that restoration practitioners often face (Grman et al., 2015).

5. Implications for ecological restoration

Our analysis provides strong evidence that mycorrhizal inoculation can promote plant growth and plant species richness in field conditions, supporting their application to ecological restoration projects. Greatest benefits would be expected in projects involving P-limited soils (e.g. mine sites and areas degraded by soil erosion) or where nitrogen-fixing woody species are planted to initiate ecological restoration (e.g., Daws et al., 2015). Our finding that C4-grasses respond more strongly to mycorrhizal inoculation could be applied to the ecological restoration of highly degraded sites where the goal is to rapidly establish ground cover. From a restoration practice perspective, our analysis highlights that whole soil inoculum could be the most effective method for achieving inoculation (Barea et al., 2011; Maltz and Treseder, 2015). Finally, our conclusion that mycorrhizal inoculation could enhance plant species richness is promising, but is based on few studies. Further research is needed to assess the benefits of using mycorrhizal inoculation to restore whole plant communities, including measures of diversity and composition, over a broader range of ecosystem types and environmental conditions.

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Supplementary data

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